

Vol 11 Chapter 4 — Holomorphic Discrete Series and Highest Weights

Keeper (author pass — deep math/physics revision)

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Chapter 4 — Holomorphic Discrete Series and Highest Weights

Level 1 — one sentence

The holomorphic discrete series of unitary representations on bounded symmetric domains is the natural unitary realization on $H^2(D; L_\lambda)$ for line bundles L_λ , with the substrate-Cartan half-sum of positive roots $\rho = (5/2, 3/2)$ providing the specific BST-relevant parameters that propagate through mass formulas and substrate operator zoo Casimir eigenvalues.

Level 2 — graduate-physicist precision

4.1 Holomorphic discrete series

For a non-compact Hermitian symmetric space G/K (with G semisimple, K maximal compact), the holomorphic discrete series is the family of unitary irreducible representations realized on $H^2(D; L_\lambda)$ — holomorphic L^2 sections of line bundles L_λ over the bounded domain $D = G/K$.

Parameterized by Harish-Chandra parameters λ . The lowest member is the Bergman representation (on $H^2(D)$ itself, $\mathbb{Q}$ = $\mathbb{R}$ Bergman normalization).

4.2 Highest-weight modules

For semisimple Lie algebra $\mathfrak{g}$ with Cartan subalgebra $\mathfrak{h}$: irreducible highest-weight modules $V(\lambda)$ parameterized by highest weight $\lambda \in \mathfrak{h}^*$.

Verma module $M(\lambda)$ — universal highest-weight module. $V(\lambda) = M(\lambda)/\text{radical}$.

For compact $\mathfrak{g}$: $V(\lambda)$ finite-dimensional, classified by integral dominant weights (Cartan-Weyl).

For non-compact: discrete-series representations are highest-weight (or lowest-weight) with parameters in specific cones.

4.3 $\rho = (5/2, 3/2)$ for BST

For $\mathfrak{so}(5, 2)$ Cartan subalgebra (rank 2): positive roots and their half-sum ρ are:

Choosing standard positive roots: $\rho = (5/2, 3/2)$.

These specific values appear throughout BST: $-\rho_2 = 3/2 - 1 = 1/2$ — zero-point energy shift of harmonic oscillator (Vol 5 Ch 2 Section 2.7) - $\rho_1 + \rho_2 = 4$ — rank-2 substrate symmetry budget - $\rho_1 \cdot \rho_2 = 15/4$ — appearing in various Casimir combinations

4.4 Harish-Chandra parameter and weights

For Wallach representations: the Harish-Chandra parameter λ specifies which holomorphic discrete series rep. Layer index k relates to $\lambda - \rho$ via integer shift.

The substrate's first-excited K-type $(1, 1)$ has weight $\lambda - \rho = (1 - 5/2, 1 - 3/2) = (-3/2, -1/2)$. Casimir on this rep $= \lambda \cdot (\lambda - 2\rho)$ (with appropriate normalization) $= \dots = 6$ (BST primary C_2).

4.5 K-audit anchors

- Knapp, *Representation Theory of Semisimple Groups*
- Vol 5 Ch 1: substrate Hilbert space (built from holomorphic discrete series)
- Vol 5 Ch 4: Casimir on K-type $(1,1) = 6$ (BST primary)

Level 3 — 5th-grader accessibility

The holomorphic discrete series is the unitary representation family realized on holomorphic L^2 functions of a bounded symmetric domain. For BST: the substrate has Cartan parameters $\rho = (5/2, 3/2)$ — these appear in BST mass formulas, in the harmonic-oscillator zero-point shift ($\hbar\omega/2$ comes from $\rho_2 = 1/2$), and in Casimir computations. The substrate's first-excited K-type $(1,1)$ has Casimir $= 6 = C_2$, the BST primary.

What comes next

Chapter 5 develops the explicit D_{IV}^5 geometry (coset, boundary, coordinates).

Where to look this up

- Knapp, *Representation Theory of Semisimple Groups*
- BST Vol 5 Ch 1-4